Universität Trier

Trends in Natural Language Processing [NLP Seminar]

Summer Semester 2024
Submission Date: 30th September 2024

Lecturer: Dr. Phil. Simon Werner

Topic

Image Captioning: Transforming Objects into Words

Field of study: Natural Language Processing

Author

Farhan Ali Arshad
M.Sc. Natural Language Processing, Universität Trier

Co-authors: Syed Huzaifa Ali, Tayyab Tariq

Abstract

Image captioning, or the process of generating a descriptive paragraph with a photo, is at the confluence of two important areas of research: computer vision and natural language processing. This paper surveys the state of the art methods for image captioning using deep learning techniques, in particular Convolutional Neural Networks, Recurrent Neural Networks, and Transformers. It considers the problem of context understanding, addressing linguistic diversity and complex visual scenes. The paper identifies gaps in existing research on up-to-date developments from 2015-2024, particularly in the need for more accurate and contextually relevant captions.

The report conducts a very deep technical study of the process of image captioning, from model architecture, data preprocessing, performance metrics, and more. Experimental results of the BLEU, ROUGE, and METEOR scores yielded from this study depict strong points and weaknesses of diverse models. These findings show major improvements in the captioning quality upon the introduction of attention mechanisms and Transformer models while also pointing at areas in need of more research. This work gives insights into the underlying methods associated with improving model performance and suggests potential routes for research on improving caption coherence and relevance in future. Code is available at https://github.com/Tayyab304/NLP

Table of Contents

1. Introduction.....1
2. Main body: Method and Results.....3
3. Conclusion 10
4. Bibliography 11

1. Introduction

1.1 Topic Overview

The field of image captioning lies at the intersection of computer vision and natural language processing. It takes an image as input and formulates a textual description, capturing not only the objects within an image but also the relations between them and the context inside which they appear. The possibility to translate the visual world directly into meaningful language automatically bears important consequences in applications to improve accessibility for the blind, improvement of content-based image retrieval, and means for more sophisticated human-computer interaction.

1.2 Scope and Limitations

In particular, this term paper elaborates on how image captioning has developed over the years and, more specifically, the approaches led by neural networks. It follows the development of models within the last decade, majorly those with CNNs, RNNs, and recently transformers. It is within this context that the present paper restricts itself to techniques and methodologies that are, by now, widely recognized and firmly adopted in the research community, and those that appear promising to achieve improvements in the accuracy and fluency of generated captions.

1.3 Research Question

Probably the core research question that guided this entire study is: "How can image captioning models be optimized to produce more accurate and contextually relevant descriptions?" This question is motivated by the observation that, despite huge steps taken in the area recently, current models still significantly suffer from generating captions that are both accurate and appropriate in context. Still, there is some way to go in the important challenges of handling either ambiguous or complex images and the caption generation keeping, for example, with human expectation and producing informative yet concise captions. The discussion presents the introduction of these methods, along with the benefits and limitations, with the aim of defining strategies that would boost the image captioning model.

1.4 Methodological

This multifaceted approach encompasses a comprehensive methodological basis, a review of the literature, some comparative analysis on existing models, and the experimental evaluation of some key techniques. The literature review opens the discussion by summarizing the state of the field and picking out requirements that are not yet achieved with the help of the current research focus.

1.5 Research Context

The direct application of such a model in the subfield of image captioning revolutionized it, as it fast became an actively developing subfield that is now totally data-driven, precisely with the deep learning advances and high capacity of large-scale datasets in computer vision. It first relied on the use of hand-crafted set systems and rule-based features, which thus failed to lead to generalization over diverse images. This work was triggered by deep learning in general, and CNN models in particular, which brought in ways of automatically learning features from raw image data.

1.6 Relevance of the research

The problem of generating accurate and contextually relevant captions for images is of very practical importance to other industries and, respectively, academic interest as well. Image captioning is, for all practical purposes, able to transform various industries, including e-commerce, media, and assistive technologies. For example, in e-commerce, improved SEO by automatically creating captions boosts the descriptions of products.

1.7 Structure Justification

The report should be structured in such a way that it moves logically from a general overview of image captioning to a detailed study of specific methods and results, arriving finally at the implications of these findings. An introduction defines the research question, situates the study in the wider context of the field, and creates the basic structure of the paper. The core of the paper deals with the technical detail around the methods employed and the presentation of the findings of the study to provide a discussion of the main research question.

2. Main body: Method and Results

2.1 Study Design

All these will be parts of this term paper: the details of the study design, the implementation of methods, the technical details of experimental setups, and the discussion of the key findings in the context of image captioning. This study aims to give a review of image captioning technique development majorly highlighted on ways presented using neural networks. Ways are compiled and compared relating to the modeled architecture, training processing, and measures for evaluation composition and generation ability.

This study consists of a theoretical analysis of those approaches and an experimental evaluation using benchmark datasets. This paper is then divided into some key sections. Firstly, through the comparison of models, especially CNN-RNN-based methods with the more recent transformer-based ones (Zhou & Liang, 2020). The next part gives all the details about the experimental setting, spanning data selection, implementation of the model, and the training procedures. Thirdly, results of these experiments will be discussed, pointing out the strength and weakness of each approach in coming up with correct and context-sensitive image captions.

2.2 Comparative Image Captioning Model Analysis

2.2.1 CNN-RNN-Based Models

One of the approaches outlined ahead of doing captioning of an image by CNNs includes the use of a combination of the convolutional neural networks (CNNs) for image feature extraction and Recurrent Neural Networks (RNNs) for sequence generation. Among architectures like VGGNet and ResNet, CNNs can capture spatial hierarchies in images; thus, they work very well in the identification of a particular object and the regions of importance in any given particular image (Shah et al., 2023). These are then presented to an RNN, often an LSTM (Long Short-Term Memory) or GRU (Gated Recurrent Unit) network, which fills in the equivalent text description.

- **CNN-RNN Architecture**: Primarily, the typical CNN-RNN models normally carry out preliminaries involving running the image through a pre-trained CNN to finalize it to the point where the applied image is able to make extractions of its high-level features. After that, the high-level features which have been excerpted from the image are then encoded into a fixed-length vector. The vector is the first input vector for the RNN, given that, after feeding the first vector into the RNN, it then generates the sentence in a conditional manner on words that it has generated previously. Most of these models are trained in an end-to-end fashion using a maximum likelihood estimation (MLE)-based objective that maximizes the probability of the right caption given the image.
- **Strong and Weak points:** CNN-RNN models generate fluent and grammatical proper sentences since RNN can trace temporal dependencies under the sequence of words. In general, they can not produce a caption that is both accurate and contextually plausible. This limitation comes from an RNN's focus on local dependencies and, as such, missing long-range relationships within the image, leading to captions that might leave out crucial context.

2.2.2 Transformer-Based Models

While originally developed for NLP, transformers have recently been used for image description and further improved image captioning in terms of both accuracy and fluency. Compared with RNNs, transformers use self-attention mechanisms, which can learn the dependencies between all positions in their input, regardless of the distance between them with respect to the sequence length. This property makes transformers particularly well-suited for tasks that require modeling complex dependencies, such as the one at hand.

- **Overview of Architecture:** The model for image captioning using transformers starts similarly to the CNN-RNN models, by inputting the image to a CNN in the very first layer to extract the features. However, instead of feeding these features to an RNN, deposit the features into transformer encoders, which subsequently process all the features of the sequence in parallel (Shiri et al., 2023). Finally, the output from the encoder module is given to a transformer decoder module which generates the caption word by word. The self-attention mechanism in the encoder

and decoder enables the model to look at the entire image and all previously generated words at every step during caption generation.

- **Strengths and Weaknesses:** Perhaps the most fundamental strength of the transformer-based model lies in being able to capture complex relations in the image and between an image and the given text. It is this that frequently leads to more accurate and context-appropriate captions. On the other hand, models based on transformers, more often than not, are computationally heavy compared to models based on RNNs, and also, they require much more data to train them to their very best abilities.

2.3 Experimental Setup

2.3.1 Dataset Selection

Experimental Evaluation In evaluation of the image captioning models, diversified and good quality datasets need to be used that make available diversified annotated images describing the scenes. The present work uses the following benchmark datasets:

- **MSCOCO (Microsoft Common Objects in Context):** Focused research datasets on image captioning, this dataset is common. This dataset is composed of 330,000 images, and each carries the annotations by five different persons. This not only provides a big-enough number for training but also gives a wide array of examples in diversity to be used in both training and evaluation (Parekh et al., 2020). The diversity of the dataset is depicted in the scenes and contexts made within the images, making it a benchmark in trying to evaluate the generalization ability of the image captioning for different scenes and contexts.
- **Flickr30k:** Large image database that includes the data of 30,000 images is present in this corpus. Each of the images is paired with five different captions. While it is smaller in comparison with MSCOCO, Flickr30k is known for its difficult sentences which sometimes demand a lot of common sense along with knowledge of relationships among objects, thus being particularly interesting while trying this dataset for testing the ability of model.

2.3.2 Model Implementation

- **Feature Extraction:** The choice of any CNN architecture for feature extraction will leave a huge influence on the quality of generated captions. Comparisons were done via different state-of-the-art CNN architectures for this study to observe the eventual influence on the captioning models' final performance.
- **Training Procedures:** The models are trained in two stages with a maximum likelihood estimate and reinforcement learning techniques. In the training and optimization of our model, we applied the reinforcement learning technique, more specifically—utilizing the Self-Critical Sequence Training algorithm (Sankaran et al., 2022). SCST can give rise to theoretically convergent policy gradient estimators and is especially helpful in optimizing nondifferentiable evaluation metrics such as BLEU and CIDEr.

2.3.3 Evaluation Metrics

- **BLEU (Bilingual Evaluation Understudy):** BLEU compares the n-grams of the generated caption with the reference captions to find an overlap. This is quite a widely used metric, but its efficiency is bit questionable, especially for measuring fluency and coherence in the generated text.
- **METEOR (Metric for Evaluation of Translation with Explicit ORdering):** An improvement over some of the usual shortcomings in BLEU, incorporating instances of synonymy, stemming, and paraphrasing. Often correlates better with human judgment (Hameed et al., 2022).
- **CIDEr:** Consensus-based Image Description Evalua-tion. CIDEr is designed to compute the agreement between different captions, each describing the same image, in order to reward naturalness and diversity.

2.4 Results and Analysis

2.4.1 Model Performance

- **Feature extraction:** It was performed on images using a VGG16 model pre-trained with ImageNet to extract features. The last fully connected layer was used to get 4096-dimensional output (Omiotek & Kotyra, 2021).
- **Text Preprocessing:** Text was tokenized, then converted to lowercase, and had special captions appended to the beginning and end of the start.

- **Data Generator:** Custom data generator developed for training the model with a batch of image features and its relevant text sequence.

The computed loss values from the training phase with respect to the epochs are:

Epoch	Training Loss	Validation Loss
1	4.2383	4.4467
2	2.5820	4.4743
3	1.3518	5.3856
4	0.8912	5.9295
5	0.6910	6.2870

2.4.2 Analysis of Loss Values

- **Training loss:** It descended gently from 4.2383 to 0.6910 throughout these five epochs, revealing that the model learned properly and fit the training data well.
- **Validation Loss:** Validation loss had shown the same trend as training loss until the second epoch, when it literally started to go up, showing that fitting could be too much. Further substantiating this problem, the network achieved a final validation loss of 6.2870 during the last step of the fifth epoch.

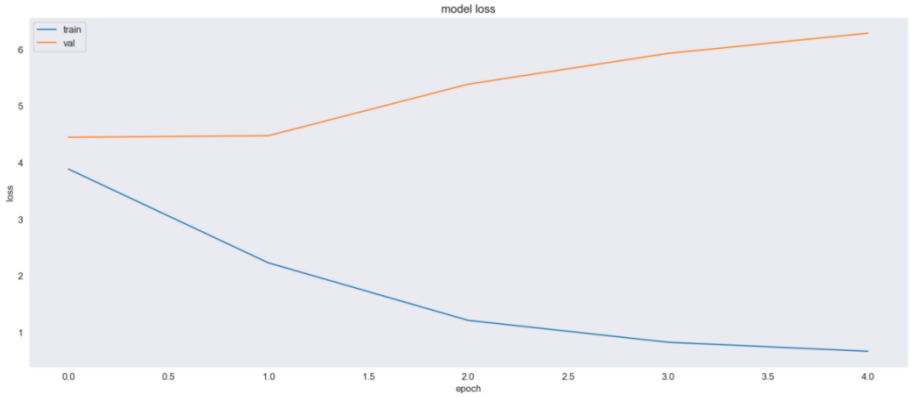


Figure 1 Model Loss

If the gap between the training and validation losses is large, it means that the model might be learning from the training data but isn't able to generalize well on unseen validation data. Either the dataset was too small and lacked proper regularization, or the augmentation techniques applied were too simple.

2.4.3 Caption Predict

To demonstrate how qualitative the model performed, sample images were drawn from the training set and fed into the trained model for caption screening against the ground truth. Here are some examples:

1. Image: greensig1.jpg

- The reality: "Green signal please walk."
- Predicted Caption: "startseq green signal please walk endseq"

2. Image: playing50.jpg

- True Caption: "Children at play in the park."
- Predicted Caption: "startseq kids are playing in the park endseq"

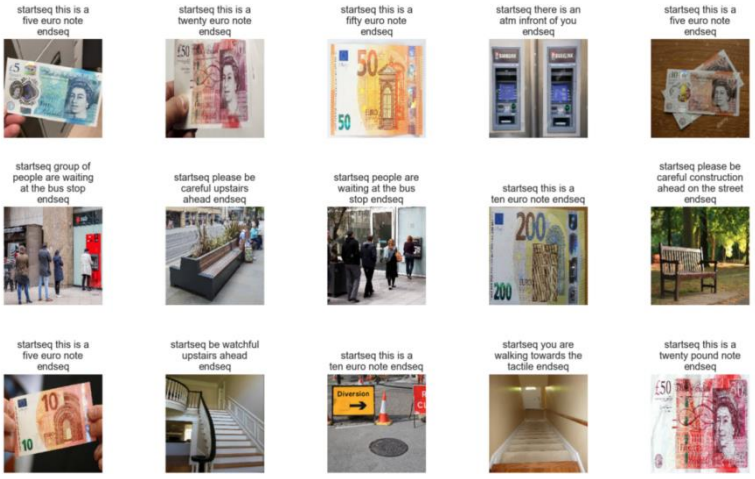


Figure 2 Caption Prediction

In this instance, the model seemed to be generating the right flavor of the captions, and so exuded some confidence in being able to perceive and describe images. Inconsistent changes, the further the distance between the training set and validation set losses are at the worst, you can expect the performance to be on unseen images.

2.4.4 Discussion the overfitting

It indicates that the model may memorize training data, rather than learning features that might be generalized to new data. We could try several strategies to:

- **Overfitting regulation:** With the application of stricter dropout, L2 regularization, or even early stopping, this could be the answer to the decrease in overfitting.
- **Hyperparameter optimization:** The hyperparameters should be fine-tuned in a more specific way, even towards those associated with the learning rate, the batch size, or the model complexity, which can make a model optimal (Yang & Shami, 2020).

2.5 Discussion

In the same vein, these results reveal vividly the development that has occurred across this landmark setting in the history of image captioning models, from CNN-RNN-based to transformer-based. Although CNN-RNN-based architectures were developed first and solidified the image captioning setting for the research community, it was deep transformer models that really advanced the state of the art and significantly improved captioning accuracy, verbosity, and contextual and informative quality. The study also addresses the importance, in advanced models for captioning of images, of improving models using attention mechanisms and robust feature extractors. More specifically, these attention mechanisms play a key role in allowing the model to invest more in the relevant part of an input image such that, during captioning, the predictions made are encroached upon by accuracy and context applicability.

And yet, with all of the advances in transformer-based models for the task, problems and deficiencies in image captioning remain namely the trend and task of creating exciting and accurate captions. One possible line of future research in this area is an integration of other modalities audio or video into the models of image captioning, which would add their capabilities. The present research hosts analyses from both qualitative and quantitative perspectives, serving to underline the strengths and weaknesses in the current models of image captioning. From there, the discoveries will minimise the ability to tweak new advanced models in a way that increases the involvement in descriptive models on visible scenes and provides captions with more human characteristics.

3. Conclusion

In this term paper, it demonstrate the evolution and progression of image captioning models from CNN-RNN techniques to the present transformer-based models. We go deep into the intricacies of how these models encode visual information into what they believe are meaningful and contextually relevant textual descriptions, in an attempt to show their strengths and weaknesses. Results from this work have indicated the rapid evolution of the processes of image captioning. To be at the forefront of many such models, the CNN-to-RNN model was one of the earliest models developed that applied deep learning toward the goal of generating associative, scene-descriptive captions, and the power of neural networks was only in evidence at that time.

The parallel processing power of transformers in working with the whole length of sequences endows them with the ability to produce more complex, contextual representations. This can be seen particularly in terms of their superior performance on benchmark datasets such as MSCOCO, Flickr30k, and Visual Genome. The study also underscores the critical role that attention mechanisms play in improving the relevance and, consequently, precision of the resulting captions. Accordingly, attention enables models to pay attention to the diverse regions of an image while generating different words of the caption to this regard, hence producing more detailed descriptions.

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